

Dataset Fairness Evaluation Report

Stage 4: Base Fairness ML Model

Detailed Group Metrics: Age

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Early-Middle	4582	0.8123	0.7714	0.3112	0.2658	0.3745	0.1033
Senior	890	0.8461	0.7704	0.2438	0.1820	0.4424	0.0609
Young	3990	0.9526	0.7329	0.0596	0.0333	0.6176	0.0112
Late-Middle	2749	0.7788	0.7584	0.3787	0.3307	0.3554	0.1393

Detailed Group Metrics: Education

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Bachelors	2071	0.8011	0.7935	0.4153	0.3930	0.2663	0.1511
9th	187	0.9679	0.4918	0.0321	0.0000	1.0000	0.0000
HS-grad	4011	0.8644	0.6985	0.1576	0.1007	0.6108	0.0468
Assoc-voc	497	0.8089	0.7353	0.2656	0.2072	0.4697	0.0904
Prof-school	224	0.8705	0.8341	0.7009	0.7679	0.0446	0.3284
10th	355	0.9380	0.6942	0.0789	0.0282	0.7143	0.0061
Some-college	2683	0.8580	0.7370	0.1826	0.1390	0.5082	0.0602
7th-8th	249	0.9197	0.5622	0.0884	0.0080	0.9091	0.0000
Masters	637	0.7928	0.7914	0.5589	0.5212	0.2191	0.1922
11th	442	0.9593	0.6895	0.0475	0.0204	0.7143	0.0071
5th-6th	99	0.9091	0.4762	0.0909	0.0000	1.0000	0.0000
Assoc-acdm	379	0.8047	0.7194	0.2427	0.2058	0.4783	0.1045
Preschool	22	1.0000	1.0000	0.0455	0.0455	0.0000	0.0000
12th	154	0.9481	0.7640	0.0844	0.0325	0.6154	0.0000

Doctorate	155	0.7226	0.6611	0.6581	0.7677	0.1275	0.5660
1st-4th	46	0.9783	0.4945	0.0217	0.0000	1.0000	0.0000

Detailed Group Metrics: Marital-status

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Never-married	4065	0.9690	0.7459	0.0440	0.0189	0.6369	0.0031
Married-civ-spouse	5567	0.7320	0.7259	0.4489	0.4026	0.3501	0.2011
Divorced	1682	0.9287	0.7200	0.0957	0.0410	0.6584	0.0092
Separated	373	0.9464	0.7585	0.0777	0.0402	0.5862	0.0087
Married-spouse-absent	152	0.9276	0.7764	0.1184	0.0592	0.5556	0.0075
Widowed	359	0.9387	0.7088	0.0864	0.0251	0.7097	0.0000
Married-AF-spouse	13	0.7692	0.7068	0.3846	0.1538	0.6000	0.0000

Detailed Group Metrics: Occupation

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Other-service	1248	0.9591	0.6303	0.0441	0.0128	0.8182	0.0050
Farming-fishing	370	0.8838	0.6953	0.1243	0.0892	0.6087	0.0463
Sales	1368	0.8341	0.7688	0.2617	0.2069	0.4218	0.0752
Exec-managerial	1472	0.7962	0.7960	0.5061	0.4654	0.2416	0.1651
Craft-repair	1564	0.7999	0.6848	0.2212	0.1746	0.5578	0.0985
Prof-specialty	1568	0.8068	0.8007	0.4298	0.3960	0.2641	0.1398
Machine-oper-inspct	727	0.8556	0.6293	0.1486	0.0702	0.7500	0.0388
Tech-support	394	0.8401	0.7973	0.3020	0.2386	0.3697	0.0691
Adm-clerical	1413	0.8946	0.7528	0.1331	0.1097	0.4840	0.0473
Missing	715	0.9189	0.7236	0.0965	0.0629	0.5942	0.0263
Handlers-cleaners	514	0.9436	0.7215	0.0700	0.0370	0.6389	0.0126
Transport-moving	576	0.8299	0.6963	0.1875	0.1493	0.5556	0.0812

Protective-serv	229	0.7686	0.7113	0.2926	0.2620	0.4478	0.1420
Priv-house-serv	52	0.9615	0.7400	0.0577	0.0192	0.6667	0.0000
Armed-Forces	1	1.0000	1.0000	0.0000	0.0000	0	0.0000

Detailed Group Metrics: Relationship

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Own-child	1904	0.9921	0.7707	0.0126	0.0047	0.6250	0.0000
Husband	4916	0.7319	0.7263	0.4530	0.4038	0.3502	0.2001
Not-in-family	3148	0.9295	0.7354	0.0985	0.0451	0.6290	0.0095
Unmarried	1311	0.9519	0.7087	0.0648	0.0214	0.7059	0.0024
Wife	580	0.7155	0.7121	0.4517	0.4397	0.3282	0.2484
Other-relative	352	0.9688	0.6685	0.0398	0.0085	0.7857	0.0000

Detailed Group Metrics: Race

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
White	10442	0.8447	0.7836	0.2547	0.2139	0.3850	0.0768
Black	1186	0.9233	0.7899	0.1214	0.0818	0.4792	0.0211
Amer-Indian-Eskimo	126	0.9127	0.7143	0.0873	0.0794	0.5455	0.0435
Asian-Pac-Islander	369	0.8347	0.7697	0.2602	0.2087	0.4167	0.0769
Other	88	0.8977	0.6720	0.1250	0.0455	0.7273	0.0130

Detailed Group Metrics: Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Female	4057	0.9258	0.7863	0.1117	0.0804	0.4724	0.0241
Male	8154	0.8169	0.7729	0.3028	0.2571	0.3779	0.0985

Detailed Group Metrics: Native-country

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
United-States	10993	0.8506	0.7847	0.2430	0.2037	0.3882	0.0727
Ireland	12	0.8333	0.8125	0.4167	0.2500	0.4000	0.0000
Cuba	30	0.8667	0.8137	0.2667	0.2000	0.3750	0.0455
Missing	209	0.8804	0.8390	0.2727	0.2201	0.3158	0.0461
Puerto-Rico	49	0.9796	0.9231	0.0816	0.0612	0.2500	0.0000
Philippines	73	0.8356	0.7349	0.2192	0.1644	0.5000	0.0702
China	32	0.8125	0.7818	0.3125	0.3125	0.3000	0.1364
Germany	54	0.8889	0.8393	0.2593	0.1852	0.3571	0.0250
Mexico	211	0.9668	0.6026	0.0284	0.0142	0.8333	0.0098
Peru	11	1.0000	1.0000	0.0909	0.0909	0.0000	0.0000
India	36	0.7500	0.7333	0.4722	0.2778	0.4706	0.0526
Guatemala	21	1.0000	1.0000	0.0000	0.0000	0	0.0000
Taiwan	15	0.8667	0.8661	0.4667	0.4667	0.1429	0.1250
Vietnam	23	0.8261	0.6167	0.0435	0.2174	0.0000	0.1818
Jamaica	26	0.8077	0.4468	0.1923	0.0000	1.0000	0.0000
Hungary	4	1.0000	1.0000	0.2500	0.2500	0.0000	0.0000
Portugal	10	0.7000	0.4118	0.3000	0.0000	1.0000	0.0000
Greece	14	0.6429	0.6410	0.5714	0.3571	0.5000	0.1667
Haiti	20	0.9000	0.8039	0.1500	0.1500	0.3333	0.0588
Nicaragua	17	1.0000	1.0000	0.0000	0.0000	0	0.0000
South	26	0.8846	0.8462	0.2692	0.2308	0.2857	0.0526
France	7	0.4286	0.4167	0.4286	0.7143	0.3333	0.7500
Canada	48	0.7292	0.6485	0.3542	0.1667	0.6471	0.0645
El-Salvador	35	0.9714	0.9206	0.1143	0.0857	0.2500	0.0000
England	38	0.7632	0.7039	0.3421	0.2105	0.5385	0.0800
Poland	18	0.7222	0.5567	0.3333	0.0556	0.8333	0.0000
Outlying-US(Guam-USVI- etc)	9	1.0000	1.0000	0.0000	0.0000	0	0.0000
Honduras	9	0.8889	0.4706	0.1111	0.0000	1.0000	0.0000
Dominican-R epublic	20	0.9500	0.8857	0.1000	0.1500	0.0000	0.0556

Italy	28	0.7857	0.7375	0.3571	0.2143	0.5000	0.0556
Trinidad&Tobago	10	1.0000	1.0000	0.1000	0.1000	0.0000	0.0000
Iran	14	0.6429	0.5238	0.2143	0.2857	0.6667	0.2727
Columbia	19	0.8947	0.7206	0.0526	0.1579	0.0000	0.1111
Hong	7	0.7143	0.4167	0.1429	0.1429	1.0000	0.1667
Yugoslavia	7	0.7143	0.4167	0.2857	0.0000	1.0000	0.0000
Ecuador	7	1.0000	1.0000	0.0000	0.0000	0	0.0000
Cambodia	7	0.7143	0.7083	0.5714	0.2857	0.5000	0.0000
Japan	23	0.9130	0.8873	0.3043	0.2174	0.2857	0.0000
Scotland	5	1.0000	1.0000	0.2000	0.2000	0.0000	0.0000
Laos	6	0.8333	0.7778	0.3333	0.1667	0.5000	0.0000
Thailand	8	1.0000	1.0000	0.0000	0.0000	0	0.0000

Stage 4.5: Per-Attribute Fairness ML Model

Detailed Group Metrics: Age

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Early-Middle	4582	0.8123	0.7714	0.3112	0.2658	0.3745	0.1033
Senior	890	0.8461	0.7704	0.2438	0.1820	0.4424	0.0609
Young	3990	0.9526	0.7329	0.0596	0.0333	0.6176	0.0112
Late-Middle	2749	0.7788	0.7584	0.3787	0.3307	0.3554	0.1393

Detailed Group Metrics: Education

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Bachelors	2071	0.8011	0.7935	0.4153	0.3930	0.2663	0.1511
9th	187	0.9679	0.4918	0.0321	0.0000	1.0000	0.0000
HS-grad	4011	0.8644	0.6985	0.1576	0.1007	0.6108	0.0468
Assoc-voc	497	0.8089	0.7353	0.2656	0.2072	0.4697	0.0904
Prof-school	224	0.8705	0.8341	0.7009	0.7679	0.0446	0.3284

10th	355	0.9380	0.6942	0.0789	0.0282	0.7143	0.0061
Some-college	2683	0.8580	0.7370	0.1826	0.1390	0.5082	0.0602
7th-8th	249	0.9197	0.5622	0.0884	0.0080	0.9091	0.0000
Masters	637	0.7928	0.7914	0.5589	0.5212	0.2191	0.1922
11th	442	0.9593	0.6895	0.0475	0.0204	0.7143	0.0071
5th-6th	99	0.9091	0.4762	0.0909	0.0000	1.0000	0.0000
Assoc-acdm	379	0.8047	0.7194	0.2427	0.2058	0.4783	0.1045
Preschool	22	1.0000	1.0000	0.0455	0.0455	0.0000	0.0000
12th	154	0.9481	0.7640	0.0844	0.0325	0.6154	0.0000
Doctorate	155	0.7226	0.6611	0.6581	0.7677	0.1275	0.5660
1st-4th	46	0.9783	0.4945	0.0217	0.0000	1.0000	0.0000

Detailed Group Metrics: Marital-status

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Never-married	4065	0.9690	0.7459	0.0440	0.0189	0.6369	0.0031
Married-civ-spouse	5567	0.7320	0.7259	0.4489	0.4026	0.3501	0.2011
Divorced	1682	0.9287	0.7200	0.0957	0.0410	0.6584	0.0092
Separated	373	0.9464	0.7585	0.0777	0.0402	0.5862	0.0087
Married-spouse-absent	152	0.9276	0.7764	0.1184	0.0592	0.5556	0.0075
Widowed	359	0.9387	0.7088	0.0864	0.0251	0.7097	0.0000
Married-AF-spouse	13	0.7692	0.7068	0.3846	0.1538	0.6000	0.0000

Detailed Group Metrics: Occupation

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Other-service	1248	0.9591	0.6303	0.0441	0.0128	0.8182	0.0050
Farming-fishing	370	0.8838	0.6953	0.1243	0.0892	0.6087	0.0463
Sales	1368	0.8341	0.7688	0.2617	0.2069	0.4218	0.0752
Exec-managerial	1472	0.7962	0.7960	0.5061	0.4654	0.2416	0.1651

Craft-repair	1564	0.7999	0.6848	0.2212	0.1746	0.5578	0.0985
Prof-specialty	1568	0.8068	0.8007	0.4298	0.3960	0.2641	0.1398
Machine-op-i nspct	727	0.8556	0.6293	0.1486	0.0702	0.7500	0.0388
Tech-support	394	0.8401	0.7973	0.3020	0.2386	0.3697	0.0691
Adm-clerical	1413	0.8946	0.7528	0.1331	0.1097	0.4840	0.0473
Missing	715	0.9189	0.7236	0.0965	0.0629	0.5942	0.0263
Handlers-clea ners	514	0.9436	0.7215	0.0700	0.0370	0.6389	0.0126
Transport-mo ving	576	0.8299	0.6963	0.1875	0.1493	0.5556	0.0812
Protective-ser v	229	0.7686	0.7113	0.2926	0.2620	0.4478	0.1420
Priv-house-se rv	52	0.9615	0.7400	0.0577	0.0192	0.6667	0.0000
Armed-Force s	1	1.0000	1.0000	0.0000	0.0000	0	0.0000

Detailed Group Metrics: Relationship

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Own-child	1904	0.9921	0.7707	0.0126	0.0047	0.6250	0.0000
Husband	4916	0.7319	0.7263	0.4530	0.4038	0.3502	0.2001
Not-in-family	3148	0.9295	0.7354	0.0985	0.0451	0.6290	0.0095
Unmarried	1311	0.9519	0.7087	0.0648	0.0214	0.7059	0.0024
Wife	580	0.7155	0.7121	0.4517	0.4397	0.3282	0.2484
Other-relative	352	0.9688	0.6685	0.0398	0.0085	0.7857	0.0000

Detailed Group Metrics: Race

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
White	10442	0.8447	0.7836	0.2547	0.2139	0.3850	0.0768
Black	1186	0.9233	0.7899	0.1214	0.0818	0.4792	0.0211
Amer-Indian-Eskimo	126	0.9127	0.7143	0.0873	0.0794	0.5455	0.0435
Asian-Pac-Isl ander	369	0.8347	0.7697	0.2602	0.2087	0.4167	0.0769

Other	88	0.8977	0.6720	0.1250	0.0455	0.7273	0.0130
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Detailed Group Metrics: Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Female	4057	0.9258	0.7863	0.1117	0.0804	0.4724	0.0241
Male	8154	0.8169	0.7729	0.3028	0.2571	0.3779	0.0985

Detailed Group Metrics: Native-country

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
United-States	10993	0.8506	0.7847	0.2430	0.2037	0.3882	0.0727
Ireland	12	0.8333	0.8125	0.4167	0.2500	0.4000	0.0000
Cuba	30	0.8667	0.8137	0.2667	0.2000	0.3750	0.0455
Missing	209	0.8804	0.8390	0.2727	0.2201	0.3158	0.0461
Puerto-Rico	49	0.9796	0.9231	0.0816	0.0612	0.2500	0.0000
Philippines	73	0.8356	0.7349	0.2192	0.1644	0.5000	0.0702
China	32	0.8125	0.7818	0.3125	0.3125	0.3000	0.1364
Germany	54	0.8889	0.8393	0.2593	0.1852	0.3571	0.0250
Mexico	211	0.9668	0.6026	0.0284	0.0142	0.8333	0.0098
Peru	11	1.0000	1.0000	0.0909	0.0909	0.0000	0.0000
India	36	0.7500	0.7333	0.4722	0.2778	0.4706	0.0526
Guatemala	21	1.0000	1.0000	0.0000	0.0000	0	0.0000
Taiwan	15	0.8667	0.8661	0.4667	0.4667	0.1429	0.1250
Vietnam	23	0.8261	0.6167	0.0435	0.2174	0.0000	0.1818
Jamaica	26	0.8077	0.4468	0.1923	0.0000	1.0000	0.0000
Hungary	4	1.0000	1.0000	0.2500	0.2500	0.0000	0.0000
Portugal	10	0.7000	0.4118	0.3000	0.0000	1.0000	0.0000
Greece	14	0.6429	0.6410	0.5714	0.3571	0.5000	0.1667
Haiti	20	0.9000	0.8039	0.1500	0.1500	0.3333	0.0588
Nicaragua	17	1.0000	1.0000	0.0000	0.0000	0	0.0000
South	26	0.8846	0.8462	0.2692	0.2308	0.2857	0.0526
France	7	0.4286	0.4167	0.4286	0.7143	0.3333	0.7500

Canada	48	0.7292	0.6485	0.3542	0.1667	0.6471	0.0645
El-Salvador	35	0.9714	0.9206	0.1143	0.0857	0.2500	0.0000
England	38	0.7632	0.7039	0.3421	0.2105	0.5385	0.0800
Poland	18	0.7222	0.5567	0.3333	0.0556	0.8333	0.0000
Outlying-US(Guam-USVI-etc)	9	1.0000	1.0000	0.0000	0.0000	0	0.0000
Honduras	9	0.8889	0.4706	0.1111	0.0000	1.0000	0.0000
Dominican-Republic	20	0.9500	0.8857	0.1000	0.1500	0.0000	0.0556
Italy	28	0.7857	0.7375	0.3571	0.2143	0.5000	0.0556
Trinidad&Tobago	10	1.0000	1.0000	0.1000	0.1000	0.0000	0.0000
Iran	14	0.6429	0.5238	0.2143	0.2857	0.6667	0.2727
Columbia	19	0.8947	0.7206	0.0526	0.1579	0.0000	0.1111
Hong	7	0.7143	0.4167	0.1429	0.1429	1.0000	0.1667
Yugoslavia	7	0.7143	0.4167	0.2857	0.0000	1.0000	0.0000
Ecuador	7	1.0000	1.0000	0.0000	0.0000	0	0.0000
Cambodia	7	0.7143	0.7083	0.5714	0.2857	0.5000	0.0000
Japan	23	0.9130	0.8873	0.3043	0.2174	0.2857	0.0000
Scotland	5	1.0000	1.0000	0.2000	0.2000	0.0000	0.0000
Laos	6	0.8333	0.7778	0.3333	0.1667	0.5000	0.0000
Thailand	8	1.0000	1.0000	0.0000	0.0000	0	0.0000

Detailed Group Metrics: Race + Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
White + Female	3261	0.9218	0.7915	0.1202	0.0892	0.4541	0.0268
Black + Male	600	0.8883	0.7759	0.1783	0.1133	0.4953	0.0284
White + Male	7181	0.8081	0.7687	0.3158	0.2715	0.3739	0.1079
Black + Female	586	0.9573	0.7838	0.0631	0.0410	0.5135	0.0109
Amer-Indian-Eskimo + Male	73	0.8904	0.6840	0.0685	0.1233	0.4000	0.0882

Asian-Pac-Islander + Female	131	0.9237	0.7707	0.1221	0.0611	0.5625	0.0087
Amer-Indian-Eskimo + Female	53	0.9245	0.7792	0.1132	0.0755	0.5000	0.0213
Asian-Pac-Islander + Male	238	0.7983	0.7664	0.3361	0.2941	0.3625	0.1203
Other + Female	26	0.9231	0.4800	0.0769	0.0000	1.0000	0.0000
Other + Male	62	0.9194	0.7852	0.1452	0.0645	0.5556	0.0000

Detailed Group Metrics: Age + Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Early-Middle + Female	1396	0.8918	0.7936	0.1726	0.1375	0.4149	0.0442
Early-Middle + Male	3186	0.7756	0.7516	0.3719	0.3176	0.3747	0.1354
Senior + Male	609	0.8095	0.7650	0.3169	0.2479	0.4093	0.0889
Young + Male	2376	0.9394	0.7304	0.0753	0.0442	0.6089	0.0159
Late-Middle + Female	766	0.8943	0.7826	0.1684	0.1149	0.4729	0.0314
Young + Female	1614	0.9703	0.7257	0.0366	0.0192	0.6441	0.0064
Senior + Female	281	0.9431	0.7847	0.0854	0.0569	0.5000	0.0156
Late-Middle + Male	1983	0.7322	0.7283	0.4599	0.4201	0.3344	0.2110

Stage 4.5: Intersectional Fairness ML Model

Detailed Group Metrics: Race + Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
White + Female	3261	0.9218	0.7915	0.1202	0.0892	0.4541	0.0268
Black + Male	600	0.8883	0.7759	0.1783	0.1133	0.4953	0.0284
White + Male	7181	0.8081	0.7687	0.3158	0.2715	0.3739	0.1079

Black + Female	586	0.9573	0.7838	0.0631	0.0410	0.5135	0.0109
Amer-Indian-Eskimo + Male	73	0.8904	0.6840	0.0685	0.1233	0.4000	0.0882
Asian-Pac-Islander + Female	131	0.9237	0.7707	0.1221	0.0611	0.5625	0.0087
Amer-Indian-Eskimo + Female	53	0.9245	0.7792	0.1132	0.0755	0.5000	0.0213
Asian-Pac-Islander + Male	238	0.7983	0.7664	0.3361	0.2941	0.3625	0.1203
Other + Female	26	0.9231	0.4800	0.0769	0.0000	1.0000	0.0000
Other + Male	62	0.9194	0.7852	0.1452	0.0645	0.5556	0.0000

Detailed Group Metrics: Age + Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Early-Middle + Female	1396	0.8918	0.7936	0.1726	0.1375	0.4149	0.0442
Early-Middle + Male	3186	0.7756	0.7516	0.3719	0.3176	0.3747	0.1354
Senior + Male	609	0.8095	0.7650	0.3169	0.2479	0.4093	0.0889
Young + Male	2376	0.9394	0.7304	0.0753	0.0442	0.6089	0.0159
Late-Middle + Female	766	0.8943	0.7826	0.1684	0.1149	0.4729	0.0314
Young + Female	1614	0.9703	0.7257	0.0366	0.0192	0.6441	0.0064
Senior + Female	281	0.9431	0.7847	0.0854	0.0569	0.5000	0.0156
Late-Middle + Male	1983	0.7322	0.7283	0.4599	0.4201	0.3344	0.2110

Stage 6: Post-Mitigation ML Model (Reweighting)

Detailed Group Metrics: Age

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
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Early-Middle	4582	0.8095	0.7667	0.3112	0.2608	0.3871	0.1017
Senior	890	0.8539	0.7808	0.2438	0.1787	0.4332	0.0535
Young	3990	0.9519	0.7162	0.0596	0.0291	0.6597	0.0093
Late-Middle	2749	0.7857	0.7656	0.3787	0.3281	0.3497	0.1317

Detailed Group Metrics: Education

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Bachelors	2071	0.8020	0.7929	0.4153	0.3747	0.2872	0.1346
9th	187	0.9679	0.4918	0.0321	0.0000	1.0000	0.0000
HS-grad	4011	0.8654	0.6997	0.1576	0.0997	0.6108	0.0456
Assoc-voc	497	0.8290	0.7632	0.2656	0.2072	0.4318	0.0767
Prof-school	224	0.8616	0.8280	0.7009	0.7411	0.0701	0.2985
10th	355	0.9324	0.6821	0.0789	0.0338	0.7143	0.0122
Some-college	2683	0.8565	0.7395	0.1826	0.1472	0.4898	0.0661
7th-8th	249	0.9197	0.5622	0.0884	0.0080	0.9091	0.0000
Masters	637	0.7928	0.7920	0.5589	0.5024	0.2360	0.1708
11th	442	0.9593	0.6895	0.0475	0.0204	0.7143	0.0071
5th-6th	99	0.9091	0.4762	0.0909	0.0000	1.0000	0.0000
Assoc-acdm	379	0.8179	0.7395	0.2427	0.2084	0.4457	0.0976
Preschool	22	1.0000	1.0000	0.0455	0.0455	0.0000	0.0000
12th	154	0.9481	0.7640	0.0844	0.0325	0.6154	0.0000
Doctorate	155	0.7032	0.6672	0.6581	0.6710	0.2157	0.4528
1st-4th	46	0.9783	0.4945	0.0217	0.0000	1.0000	0.0000

Detailed Group Metrics: Marital-status

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Never-married	4065	0.9680	0.7338	0.0440	0.0180	0.6592	0.0031
Married-civ-spouse	5567	0.7350	0.7285	0.4489	0.3955	0.3545	0.1920
Divorced	1682	0.9287	0.7200	0.0957	0.0410	0.6584	0.0092
Separated	373	0.9437	0.7290	0.0777	0.0322	0.6552	0.0058

Married-spouse-absent	152	0.9276	0.7603	0.1184	0.0461	0.6111	0.0000
Widowed	359	0.9387	0.7088	0.0864	0.0251	0.7097	0.0000
Married-AF-spouse	13	0.6154	0.3810	0.3846	0.0000	1.0000	0.0000

Detailed Group Metrics: Occupation

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Other-service	1248	0.9567	0.6513	0.0441	0.0200	0.7636	0.0101
Farming-fishing	370	0.8730	0.6811	0.1243	0.1000	0.6087	0.0586
Sales	1368	0.8348	0.7705	0.2617	0.2091	0.4162	0.0762
Exec-managerial	1472	0.7921	0.7914	0.5061	0.4355	0.2752	0.1389
Craft-repair	1564	0.8120	0.7028	0.2212	0.1726	0.5347	0.0895
Prof-specialty	1568	0.8125	0.8059	0.4298	0.3852	0.2700	0.1253
Machine-oper-inspct	727	0.8583	0.6479	0.1486	0.0784	0.7130	0.0420
Tech-support	394	0.8376	0.7910	0.3020	0.2259	0.3950	0.0618
Adm-clerical	1413	0.8910	0.7451	0.1331	0.1104	0.4947	0.0498
Missing	715	0.9231	0.7358	0.0965	0.0615	0.5797	0.0232
Handlers-cleaners	514	0.9397	0.7121	0.0700	0.0409	0.6389	0.0167
Transport-moving	576	0.8229	0.6813	0.1875	0.1458	0.5833	0.0833
Protective-serv	229	0.7729	0.7125	0.2926	0.2489	0.4627	0.1296
Priv-house-serv	52	0.9615	0.7400	0.0577	0.0192	0.6667	0.0000
Armed-Forces	1	1.0000	1.0000	0.0000	0.0000	0	0.0000

Detailed Group Metrics: Relationship

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Own-child	1904	0.9921	0.7707	0.0126	0.0047	0.6250	0.0000
Husband	4916	0.7333	0.7274	0.4530	0.3999	0.3529	0.1952

Not-in-family	3148	0.9285	0.7280	0.0985	0.0429	0.6452	0.0088
Unmarried	1311	0.9504	0.6943	0.0648	0.0198	0.7294	0.0024
Wife	580	0.7293	0.7234	0.4517	0.4017	0.3550	0.2013
Other-relative	352	0.9688	0.6685	0.0398	0.0085	0.7857	0.0000

Detailed Group Metrics: Race

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
White	10442	0.8452	0.7829	0.2547	0.2095	0.3925	0.0735
Black	1186	0.9207	0.7788	0.1214	0.0776	0.5069	0.0202
Amer-Indian-Eskimo	126	0.9286	0.7438	0.0873	0.0635	0.5455	0.0261
Asian-Pac-Islander	369	0.8428	0.7819	0.2602	0.2114	0.3958	0.0733
Other	88	0.9205	0.7721	0.1250	0.0682	0.5455	0.0130

Detailed Group Metrics: Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Female	4057	0.9273	0.7846	0.1117	0.0744	0.4923	0.0200
Male	8154	0.8174	0.7727	0.3028	0.2539	0.3823	0.0959

Detailed Group Metrics: Native-country

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
United-States	10993	0.8518	0.7851	0.2430	0.1998	0.3939	0.0693
Ireland	12	0.9167	0.9111	0.4167	0.3333	0.2000	0.0000
Cuba	30	0.8667	0.8137	0.2667	0.2000	0.3750	0.0455
Missing	209	0.8660	0.8208	0.2727	0.2249	0.3333	0.0592
Puerto-Rico	49	0.9592	0.8225	0.0816	0.0408	0.5000	0.0000
Philippines	73	0.8493	0.7748	0.2192	0.2055	0.3750	0.0877
China	32	0.8750	0.8333	0.3125	0.1875	0.4000	0.0000
Germany	54	0.8704	0.8272	0.2593	0.2407	0.2857	0.0750
Mexico	211	0.9716	0.6927	0.0284	0.0190	0.6667	0.0098

Peru	11	1.0000	1.0000	0.0909	0.0909	0.0000	0.0000
India	36	0.7222	0.7078	0.4722	0.3056	0.4706	0.1053
Guatemala	21	1.0000	1.0000	0.0000	0.0000	0	0.0000
Taiwan	15	0.8000	0.7964	0.4667	0.4000	0.2857	0.1250
Vietnam	23	0.8696	0.6634	0.0435	0.1739	0.0000	0.1364
Jamaica	26	0.8077	0.4468	0.1923	0.0000	1.0000	0.0000
Hungary	4	0.7500	0.4286	0.2500	0.0000	1.0000	0.0000
Portugal	10	0.7000	0.4118	0.3000	0.0000	1.0000	0.0000
Greece	14	0.5000	0.4974	0.5714	0.3571	0.6250	0.3333
Haiti	20	0.9000	0.7222	0.1500	0.0500	0.6667	0.0000
Nicaragua	17	1.0000	1.0000	0.0000	0.0000	0	0.0000
South	26	0.9231	0.8917	0.2692	0.1923	0.2857	0.0000
France	7	0.4286	0.4167	0.4286	0.7143	0.3333	0.7500
Canada	48	0.7292	0.6284	0.3542	0.1250	0.7059	0.0323
El-Salvador	35	0.9714	0.9206	0.1143	0.0857	0.2500	0.0000
England	38	0.7895	0.7286	0.3421	0.1842	0.5385	0.0400
Poland	18	0.7222	0.5567	0.3333	0.0556	0.8333	0.0000
Outlying-US(Guam-USVI-e tc)	9	1.0000	1.0000	0.0000	0.0000	0	0.0000
Honduras	9	0.8889	0.4706	0.1111	0.0000	1.0000	0.0000
Dominican-R epublic	20	0.9000	0.7222	0.1000	0.1000	0.5000	0.0556
Italy	28	0.7500	0.6813	0.3571	0.1786	0.6000	0.0556
Trinidad&To bago	10	1.0000	1.0000	0.1000	0.1000	0.0000	0.0000
Iran	14	0.7143	0.5758	0.2143	0.2143	0.6667	0.1818
Columbia	19	0.9474	0.8190	0.0526	0.1053	0.0000	0.0556
Hong	7	0.7143	0.4167	0.1429	0.1429	1.0000	0.1667
Yugoslavia	7	0.7143	0.4167	0.2857	0.0000	1.0000	0.0000
Ecuador	7	1.0000	1.0000	0.0000	0.0000	0	0.0000
Cambodia	7	0.7143	0.7083	0.5714	0.2857	0.5000	0.0000
Japan	23	0.8696	0.8392	0.3043	0.2609	0.2857	0.0625
Scotland	5	1.0000	1.0000	0.2000	0.2000	0.0000	0.0000

Laos	6	0.8333	0.7778	0.3333	0.1667	0.5000	0.0000
Thailand	8	1.0000	1.0000	0.0000	0.0000	0	0.0000

Detailed Group Metrics: Race + Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
White + Female	3261	0.9218	0.7859	0.1202	0.0831	0.4796	0.0234
Black + Male	600	0.8833	0.7691	0.1783	0.1183	0.4953	0.0345
White + Male	7181	0.8105	0.7705	0.3158	0.2670	0.3774	0.1028
Black + Female	586	0.9590	0.7823	0.0631	0.0358	0.5405	0.0073
Amer-Indian-Eskimo + Male	73	0.9315	0.7542	0.0685	0.0822	0.4000	0.0441
Asian-Pac-Islander + Female	131	0.9160	0.7379	0.1221	0.0534	0.6250	0.0087
Amer-Indian-Eskimo + Female	53	0.9245	0.7296	0.1132	0.0377	0.6667	0.0000
Asian-Pac-Islander + Male	238	0.8025	0.7721	0.3361	0.2983	0.3500	0.1203
Other + Female	26	0.9615	0.8231	0.0769	0.0385	0.5000	0.0000
Other + Male	62	0.9032	0.7584	0.1452	0.0806	0.5556	0.0189

Detailed Group Metrics: Age + Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Early-Middle + Female	1396	0.8875	0.7830	0.1726	0.1332	0.4398	0.0442
Early-Middle + Male	3186	0.7753	0.7511	0.3719	0.3167	0.3764	0.1349
Senior + Male	609	0.8062	0.7584	0.3169	0.2381	0.4301	0.0841
Young + Male	2376	0.9386	0.7133	0.0753	0.0383	0.6536	0.0132
Late-Middle + Female	766	0.8956	0.7757	0.1684	0.1005	0.5116	0.0220
Young + Female	1614	0.9715	0.7189	0.0366	0.0155	0.6780	0.0039

Senior + Female	281	0.9573	0.8307	0.0854	0.0498	0.4583	0.0039
Late-Middle + Male	1983	0.7433	0.7393	0.4599	0.4160	0.3268	0.1970

Comparative Group Metrics: Before vs After (Reweighting)

Age

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Early-Middle	0.7714	0.7667	0.3745	0.3871	0.1033	0.1017	0.2658	0.2608
Senior	0.7704	0.7808	0.4424	0.4332	0.0609	0.0535	0.1820	0.1787
Young	0.7329	0.7162	0.6176	0.6597	0.0112	0.0093	0.0333	0.0291
Late-Middle	0.7584	0.7656	0.3554	0.3497	0.1393	0.1317	0.3307	0.3281

Education

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Bachelors	0.7935	0.7929	0.2663	0.2872	0.1511	0.1346	0.3930	0.3747
9th	0.4918	0.4918	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
HS-grad	0.6985	0.6997	0.6108	0.6108	0.0468	0.0456	0.1007	0.0997
Assoc-voc	0.7353	0.7632	0.4697	0.4318	0.0904	0.0767	0.2072	0.2072
Prof-school	0.8341	0.8280	0.0446	0.0701	0.3284	0.2985	0.7679	0.7411
10th	0.6942	0.6821	0.7143	0.7143	0.0061	0.0122	0.0282	0.0338
Some-college	0.7370	0.7395	0.5082	0.4898	0.0602	0.0661	0.1390	0.1472
7th-8th	0.5622	0.5622	0.9091	0.9091	0.0000	0.0000	0.0080	0.0080
Masters	0.7914	0.7920	0.2191	0.2360	0.1922	0.1708	0.5212	0.5024
11th	0.6895	0.6895	0.7143	0.7143	0.0071	0.0071	0.0204	0.0204
5th-6th	0.4762	0.4762	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
Assoc-acdm	0.7194	0.7395	0.4783	0.4457	0.1045	0.0976	0.2058	0.2084
Preschool	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000	0.0455	0.0455
12th	0.7640	0.7640	0.6154	0.6154	0.0000	0.0000	0.0325	0.0325
Doctorate	0.6611	0.6672	0.1275	0.2157	0.5660	0.4528	0.7677	0.6710
1st-4th	0.4945	0.4945	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000

Marital-status

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Never-married	0.7459	0.7338	0.6369	0.6592	0.0031	0.0031	0.0189	0.0180
Married-civil-spouse	0.7259	0.7285	0.3501	0.3545	0.2011	0.1920	0.4026	0.3955
Divorced	0.7200	0.7200	0.6584	0.6584	0.0092	0.0092	0.0410	0.0410
Separated	0.7585	0.7290	0.5862	0.6552	0.0087	0.0058	0.0402	0.0322
Married-spouse-absent	0.7764	0.7603	0.5556	0.6111	0.0075	0.0000	0.0592	0.0461
Widowed	0.7088	0.7088	0.7097	0.7097	0.0000	0.0000	0.0251	0.0251
Married-AF-spouse	0.7068	0.3810	0.6000	1.0000	0.0000	0.0000	0.1538	0.0000

Occupation

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Other-service	0.6303	0.6513	0.8182	0.7636	0.0050	0.0101	0.0128	0.0200
Farming-fishing	0.6953	0.6811	0.6087	0.6087	0.0463	0.0586	0.0892	0.1000
Sales	0.7688	0.7705	0.4218	0.4162	0.0752	0.0762	0.2069	0.2091
Exec-managerial	0.7960	0.7914	0.2416	0.2752	0.1651	0.1389	0.4654	0.4355
Craft-repair	0.6848	0.7028	0.5578	0.5347	0.0985	0.0895	0.1746	0.1726
Prof-specialty	0.8007	0.8059	0.2641	0.2700	0.1398	0.1253	0.3960	0.3852
Machine-operator-inspct	0.6293	0.6479	0.7500	0.7130	0.0388	0.0420	0.0702	0.0784
Tech-support	0.7973	0.7910	0.3697	0.3950	0.0691	0.0618	0.2386	0.2259
Adm-clerical	0.7528	0.7451	0.4840	0.4947	0.0473	0.0498	0.1097	0.1104
Missing	0.7236	0.7358	0.5942	0.5797	0.0263	0.0232	0.0629	0.0615
Handlers-cleaners	0.7215	0.7121	0.6389	0.6389	0.0126	0.0167	0.0370	0.0409
Transport-moving	0.6963	0.6813	0.5556	0.5833	0.0812	0.0833	0.1493	0.1458
Protective-service	0.7113	0.7125	0.4478	0.4627	0.1420	0.1296	0.2620	0.2489

Priv-house-serv	0.7400	0.7400	0.6667	0.6667	0.0000	0.0000	0.0192	0.0192
Armed-Forces	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000

Relationship

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Own-child	0.7707	0.7707	0.6250	0.6250	0.0000	0.0000	0.0047	0.0047
Husband	0.7263	0.7274	0.3502	0.3529	0.2001	0.1952	0.4038	0.3999
Not-in-family	0.7354	0.7280	0.6290	0.6452	0.0095	0.0088	0.0451	0.0429
Unmarried	0.7087	0.6943	0.7059	0.7294	0.0024	0.0024	0.0214	0.0198
Wife	0.7121	0.7234	0.3282	0.3550	0.2484	0.2013	0.4397	0.4017
Other-relative	0.6685	0.6685	0.7857	0.7857	0.0000	0.0000	0.0085	0.0085

Race

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
White	0.7836	0.7829	0.3850	0.3925	0.0768	0.0735	0.2139	0.2095
Black	0.7899	0.7788	0.4792	0.5069	0.0211	0.0202	0.0818	0.0776
Amer-Indian-Eskimo	0.7143	0.7438	0.5455	0.5455	0.0435	0.0261	0.0794	0.0635
Asian-Pac-Islander	0.7697	0.7819	0.4167	0.3958	0.0769	0.0733	0.2087	0.2114
Other	0.6720	0.7721	0.7273	0.5455	0.0130	0.0130	0.0455	0.0682

Sex

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Female	0.7863	0.7846	0.4724	0.4923	0.0241	0.0200	0.0804	0.0744
Male	0.7729	0.7727	0.3779	0.3823	0.0985	0.0959	0.2571	0.2539

Native-country

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
United-States	0.7847	0.7851	0.3882	0.3939	0.0727	0.0693	0.2037	0.1998

Ireland	0.8125	0.9111	0.4000	0.2000	0.0000	0.0000	0.2500	0.3333
Cuba	0.8137	0.8137	0.3750	0.3750	0.0455	0.0455	0.2000	0.2000
Missing	0.8390	0.8208	0.3158	0.3333	0.0461	0.0592	0.2201	0.2249
Puerto-Rico	0.9231	0.8225	0.2500	0.5000	0.0000	0.0000	0.0612	0.0408
Philippines	0.7349	0.7748	0.5000	0.3750	0.0702	0.0877	0.1644	0.2055
China	0.7818	0.8333	0.3000	0.4000	0.1364	0.0000	0.3125	0.1875
Germany	0.8393	0.8272	0.3571	0.2857	0.0250	0.0750	0.1852	0.2407
Mexico	0.6026	0.6927	0.8333	0.6667	0.0098	0.0098	0.0142	0.0190
Peru	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000	0.0909	0.0909
India	0.7333	0.7078	0.4706	0.4706	0.0526	0.1053	0.2778	0.3056
Guatemala	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000
Taiwan	0.8661	0.7964	0.1429	0.2857	0.1250	0.1250	0.4667	0.4000
Vietnam	0.6167	0.6634	0.0000	0.0000	0.1818	0.1364	0.2174	0.1739
Jamaica	0.4468	0.4468	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
Hungary	1.0000	0.4286	0.0000	1.0000	0.0000	0.0000	0.2500	0.0000
Portugal	0.4118	0.4118	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
Greece	0.6410	0.4974	0.5000	0.6250	0.1667	0.3333	0.3571	0.3571
Haiti	0.8039	0.7222	0.3333	0.6667	0.0588	0.0000	0.1500	0.0500
Nicaragua	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000
South	0.8462	0.8917	0.2857	0.2857	0.0526	0.0000	0.2308	0.1923
France	0.4167	0.4167	0.3333	0.3333	0.7500	0.7500	0.7143	0.7143
Canada	0.6485	0.6284	0.6471	0.7059	0.0645	0.0323	0.1667	0.1250
El-Salvador	0.9206	0.9206	0.2500	0.2500	0.0000	0.0000	0.0857	0.0857
England	0.7039	0.7286	0.5385	0.5385	0.0800	0.0400	0.2105	0.1842
Poland	0.5567	0.5567	0.8333	0.8333	0.0000	0.0000	0.0556	0.0556
Outlying-US (Guam-USV I-etc)	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000
Honduras	0.4706	0.4706	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
Dominican- Republic	0.8857	0.7222	0.0000	0.5000	0.0556	0.0556	0.1500	0.1000
Italy	0.7375	0.6813	0.5000	0.6000	0.0556	0.0556	0.2143	0.1786
Trinidad&T obago	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000	0.1000	0.1000

Iran	0.5238	0.5758	0.6667	0.6667	0.2727	0.1818	0.2857	0.2143
Columbia	0.7206	0.8190	0.0000	0.0000	0.1111	0.0556	0.1579	0.1053
Hong	0.4167	0.4167	1.0000	1.0000	0.1667	0.1667	0.1429	0.1429
Yugoslavia	0.4167	0.4167	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
Ecuador	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000
Cambodia	0.7083	0.7083	0.5000	0.5000	0.0000	0.0000	0.2857	0.2857
Japan	0.8873	0.8392	0.2857	0.2857	0.0000	0.0625	0.2174	0.2609
Scotland	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000	0.2000	0.2000
Laos	0.7778	0.7778	0.5000	0.5000	0.0000	0.0000	0.1667	0.1667
Thailand	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000

Stage 6: Post-Mitigation ML Model (SMOTE)

Detailed Group Metrics: Age

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Late-Middle	4499	0.8526	0.8432	0.6204	0.6250	0.1150	0.2002
Early-Middle	8636	0.8908	0.8817	0.6346	0.6435	0.0790	0.1616
Young	4063	0.9503	0.7902	0.0765	0.0500	0.4984	0.0125
Senior	1380	0.8877	0.8877	0.5123	0.4855	0.1358	0.0877

Detailed Group Metrics: Education

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
HS-grad	5209	0.8798	0.8664	0.3513	0.3321	0.1984	0.0778
Bachelors	3528	0.8852	0.8704	0.6567	0.6808	0.0691	0.2023
Some-college	2920	0.8558	0.8027	0.2490	0.2322	0.3232	0.0848
11th	464	0.9504	0.8150	0.0927	0.0517	0.4884	0.0048
Assoc-voc	953	0.8877	0.8798	0.6170	0.6390	0.0731	0.1753
Masters	1215	0.8872	0.8392	0.7687	0.7778	0.0675	0.2633
Prof-school	700	0.9443	0.8262	0.9043	0.9200	0.0221	0.3731

Assoc-acdm	598	0.8562	0.8549	0.5201	0.5736	0.0868	0.2056
Preschool	546	0.9963	0.9763	0.9615	0.9579	0.0038	0.0000
5th-6th	134	0.9478	0.9389	0.3284	0.2910	0.1364	0.0111
Doctorate	1178	0.9669	0.7620	0.9550	0.9728	0.0080	0.5660
12th	183	0.9344	0.9004	0.2295	0.1858	0.2381	0.0142
9th	245	0.9265	0.8958	0.2612	0.1959	0.2656	0.0055
1st-4th	71	0.9718	0.9691	0.3662	0.3380	0.0769	0.0000
7th-8th	280	0.9286	0.8699	0.1893	0.1393	0.3208	0.0132
10th	354	0.9463	0.7540	0.0763	0.0395	0.5926	0.0092

Detailed Group Metrics: Marital-status

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Married-civ-s pouse	10737	0.8525	0.8163	0.7143	0.7295	0.0926	0.2849
Divorced	2043	0.9305	0.9036	0.2555	0.2163	0.2126	0.0204
Never-marrie d	4176	0.9638	0.8301	0.0694	0.0433	0.4483	0.0054
Married-spou se-absent	519	0.9326	0.9090	0.7418	0.7669	0.0286	0.1791
Widowed	357	0.9356	0.7024	0.0812	0.0336	0.6897	0.0091
Separated	408	0.9436	0.8836	0.1569	0.1250	0.2812	0.0145
Married-AF-s pouse	338	0.9852	0.8771	0.9763	0.9615	0.0152	0.0000

Detailed Group Metrics: Occupation

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Prof-specialty	2507	0.8803	0.8689	0.6434	0.6522	0.0862	0.1801
Machine-op-i nspct	1190	0.9151	0.9150	0.4798	0.4807	0.0876	0.0824
Handlers-clea ners	977	0.9560	0.9560	0.5107	0.5077	0.0461	0.0418
Missing	788	0.9188	0.8602	0.1802	0.1726	0.2465	0.0449
Protective-ser v	714	0.8992	0.8563	0.7731	0.7731	0.0652	0.2222

Priv-house-serv	510	0.9941	0.9832	0.9039	0.9020	0.0043	0.0204
Exec-managerial	2323	0.8687	0.8469	0.6870	0.6901	0.0934	0.2146
Other-service	1694	0.9610	0.9532	0.2957	0.2957	0.0659	0.0277
Farming-fishing	835	0.9210	0.9166	0.6120	0.6168	0.0607	0.1080
Adm-clerical	1678	0.9076	0.8839	0.2700	0.2777	0.1567	0.0686
Armed-Forces	195	0.9538	0.5790	0.9949	0.9487	0.0464	0.0000
Craft-repair	2337	0.8395	0.8391	0.4788	0.4673	0.1796	0.1429
Sales	1695	0.8690	0.8638	0.4041	0.4006	0.1664	0.1069
Tech-support	521	0.8868	0.8863	0.4722	0.4626	0.1301	0.0982
Transport-moving	614	0.7997	0.6956	0.2378	0.1775	0.5479	0.0919

Detailed Group Metrics: Relationship

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Husband	9925	0.8573	0.8162	0.7291	0.7438	0.0878	0.2904
Wife	668	0.7695	0.7686	0.5240	0.5359	0.2086	0.2547
Other-relative	655	0.9527	0.9525	0.4840	0.4611	0.0726	0.0237
Unmarried	1521	0.9645	0.9412	0.1940	0.1769	0.1356	0.0114
Not-in-family	3615	0.9225	0.8765	0.2149	0.1746	0.2741	0.0236
Own-child	2194	0.9872	0.9735	0.1431	0.1367	0.0669	0.0037

Detailed Group Metrics: Race

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
White	15671	0.8871	0.8870	0.5034	0.5020	0.1136	0.1123
Black	1633	0.9424	0.9374	0.3619	0.3546	0.0897	0.0393
Other	553	0.9747	0.9489	0.8608	0.8499	0.0210	0.0519
Asian-Pac-Islander	563	0.8739	0.8739	0.5151	0.5027	0.1345	0.1172
Amer-Indian-Eskimo	158	0.9114	0.8882	0.2722	0.2722	0.1628	0.0609

Detailed Group Metrics: Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Male	12865	0.8732	0.8714	0.5581	0.5596	0.1123	0.1451
Female	5713	0.9419	0.9372	0.3692	0.3576	0.0944	0.0369

Detailed Group Metrics: Native-country

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
United-States	16321	0.8899	0.8898	0.4901	0.4891	0.1134	0.1069
Puerto-Rico	100	0.9500	0.9496	0.5500	0.5400	0.0545	0.0444
Yugoslavia	6	0.8333	0.4545	0.1667	0.0000	1.0000	0.0000
Iran	45	0.8444	0.7826	0.7556	0.7778	0.0882	0.3636
Germany	80	0.8625	0.8623	0.5000	0.4625	0.1750	0.1000
Cuba	37	0.8919	0.8879	0.4054	0.4054	0.1333	0.0909
Trinidad&Tobago	87	0.9655	0.9114	0.8966	0.8851	0.0256	0.1111
Missing	219	0.8447	0.8140	0.3059	0.2877	0.2836	0.0987
Greece	28	0.8571	0.7879	0.7857	0.7857	0.0909	0.3333
Poland	63	1.0000	1.0000	0.8095	0.8095	0.0000	0.0000
Laos	39	0.9744	0.9372	0.8974	0.8718	0.0286	0.0000
South	69	0.9710	0.9637	0.7246	0.7246	0.0200	0.0526
Dominican-Republic	32	0.9375	0.9365	0.4375	0.4375	0.0714	0.0556
Haiti	49	0.9388	0.9333	0.6531	0.6327	0.0625	0.0588
Honduras	27	1.0000	1.0000	0.7037	0.7037	0.0000	0.0000
Mexico	252	0.9643	0.9397	0.1865	0.1746	0.1277	0.0146
Jamaica	56	0.9643	0.9626	0.6250	0.5893	0.0571	0.0000
Taiwan	74	0.9595	0.9003	0.8919	0.8784	0.0303	0.1250
Canada	60	0.9167	0.9165	0.4833	0.4667	0.1034	0.0645
Columbia	32	0.9688	0.9685	0.4375	0.4688	0.0000	0.0556
China	37	0.8378	0.8348	0.4054	0.4595	0.1333	0.1818
Peru	47	1.0000	1.0000	0.7872	0.7872	0.0000	0.0000
England	50	0.8400	0.8390	0.5000	0.4200	0.2400	0.0800

Portugal	56	0.9464	0.8960	0.8750	0.8214	0.0612	0.0000
Philippines	120	0.8833	0.8830	0.5250	0.5250	0.1111	0.1228
Vietnam	29	0.8966	0.8752	0.2414	0.3448	0.0000	0.1364
Outlying-US(Guam-USVI-e tc)	34	1.0000	1.0000	0.7353	0.7353	0.0000	0.0000
Scotland	56	0.9821	0.9396	0.9286	0.9107	0.0192	0.0000
Nicaragua	46	0.9565	0.9544	0.6304	0.5870	0.0690	0.0000
Ireland	34	0.9118	0.8717	0.7941	0.7647	0.0741	0.1429
Hungary	23	0.9565	0.9158	0.8696	0.8261	0.0500	0.0000
Italy	45	0.8889	0.8853	0.6000	0.5778	0.1111	0.1111
India	55	0.8364	0.8250	0.6545	0.6000	0.1667	0.1579
Thailand	68	0.9853	0.9625	0.8824	0.8971	0.0000	0.1250
El-Salvador	42	0.9524	0.9344	0.2619	0.2143	0.1818	0.0000
Guatemala	34	1.0000	1.0000	0.3824	0.3824	0.0000	0.0000
Japan	45	0.9333	0.9282	0.6444	0.6222	0.0690	0.0625
Hong	22	0.9545	0.9394	0.7273	0.7727	0.0000	0.1667
France	19	0.7895	0.6042	0.7895	0.8947	0.0667	0.7500
Holand-Nethe rlands	16	0.9375	0.4839	1.0000	0.9375	0.0625	0
Ecuador	12	1.0000	1.0000	0.4167	0.4167	0.0000	0.0000
Cambodia	12	0.8333	0.8125	0.7500	0.5833	0.2222	0.0000

Detailed Group Metrics: Race + Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
White + Male	11116	0.8667	0.8648	0.5580	0.5605	0.1172	0.1537
White + Female	4555	0.9368	0.9318	0.3701	0.3592	0.1002	0.0415
Black + Male	898	0.9220	0.9212	0.4510	0.4443	0.0938	0.0649
Other + Male	378	0.9656	0.9314	0.8598	0.8466	0.0277	0.0755
Asian-Pac-Isl ander + Female	186	0.9194	0.9128	0.3817	0.3441	0.1549	0.0348
Asian-Pac-Isl ander + Male	377	0.8515	0.8475	0.5809	0.5809	0.1279	0.1772

Black + Female	735	0.9673	0.9563	0.2531	0.2449	0.0806	0.0164
Amer-Indian-Eskimo + Female	62	0.9355	0.9077	0.2419	0.2097	0.2000	0.0213
Other + Female	175	0.9943	0.9881	0.8629	0.8571	0.0066	0.0000
Amer-Indian-Eskimo + Male	96	0.8958	0.8765	0.2917	0.3125	0.1429	0.0882

Detailed Group Metrics: Age + Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Late-Middle + Male	3313	0.8292	0.8025	0.6767	0.6906	0.1160	0.2857
Early-Middle + Male	6177	0.8745	0.8553	0.6761	0.6885	0.0836	0.2129
Early-Middle + Female	2459	0.9317	0.9314	0.5303	0.5303	0.0644	0.0727
Young + Female	1629	0.9724	0.8038	0.0454	0.0276	0.5000	0.0051
Late-Middle + Female	1186	0.9182	0.9175	0.4629	0.4418	0.1111	0.0565
Senior + Male	941	0.8587	0.8575	0.5579	0.5313	0.1505	0.1298
Young + Male	2434	0.9355	0.7837	0.0974	0.0649	0.4979	0.0178
Senior + Female	439	0.9499	0.9478	0.4146	0.3872	0.0934	0.0195

Comparative Group Metrics: Before vs After (SMOTE)

Age

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Early-Middle	0.7714	0.8817	0.3745	0.0790	0.1033	0.1616	0.2658	0.6435
Senior	0.7704	0.8877	0.4424	0.1358	0.0609	0.0877	0.1820	0.4855
Young	0.7329	0.7902	0.6176	0.4984	0.0112	0.0125	0.0333	0.0500
Late-Middle	0.7584	0.8432	0.3554	0.1150	0.1393	0.2002	0.3307	0.6250

Education

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Bachelors	0.7935	0.8704	0.2663	0.0691	0.1511	0.2023	0.3930	0.6808
9th	0.4918	0.8958	1.0000	0.2656	0.0000	0.0055	0.0000	0.1959
HS-grad	0.6985	0.8664	0.6108	0.1984	0.0468	0.0778	0.1007	0.3321
Assoc-voc	0.7353	0.8798	0.4697	0.0731	0.0904	0.1753	0.2072	0.6390
Prof-school	0.8341	0.8262	0.0446	0.0221	0.3284	0.3731	0.7679	0.9200
10th	0.6942	0.7540	0.7143	0.5926	0.0061	0.0092	0.0282	0.0395
Some-college	0.7370	0.8027	0.5082	0.3232	0.0602	0.0848	0.1390	0.2322
7th-8th	0.5622	0.8699	0.9091	0.3208	0.0000	0.0132	0.0080	0.1393
Masters	0.7914	0.8392	0.2191	0.0675	0.1922	0.2633	0.5212	0.7778
11th	0.6895	0.8150	0.7143	0.4884	0.0071	0.0048	0.0204	0.0517
5th-6th	0.4762	0.9389	1.0000	0.1364	0.0000	0.0111	0.0000	0.2910
Assoc-acdm	0.7194	0.8549	0.4783	0.0868	0.1045	0.2056	0.2058	0.5736
Preschool	1.0000	0.9763	0.0000	0.0038	0.0000	0.0000	0.0455	0.9579
12th	0.7640	0.9004	0.6154	0.2381	0.0000	0.0142	0.0325	0.1858
Doctorate	0.6611	0.7620	0.1275	0.0080	0.5660	0.5660	0.7677	0.9728
1st-4th	0.4945	0.9691	1.0000	0.0769	0.0000	0.0000	0.0000	0.3380

Marital-status

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Never-married	0.7459	0.8301	0.6369	0.4483	0.0031	0.0054	0.0189	0.0433
Married-civ-spouse	0.7259	0.8163	0.3501	0.0926	0.2011	0.2849	0.4026	0.7295
Divorced	0.7200	0.9036	0.6584	0.2126	0.0092	0.0204	0.0410	0.2163
Separated	0.7585	0.8836	0.5862	0.2812	0.0087	0.0145	0.0402	0.1250
Married-spouse-absent	0.7764	0.9090	0.5556	0.0286	0.0075	0.1791	0.0592	0.7669
Widowed	0.7088	0.7024	0.7097	0.6897	0.0000	0.0091	0.0251	0.0336
Married-AF-spouse	0.7068	0.8771	0.6000	0.0152	0.0000	0.0000	0.1538	0.9615

Occupation

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Other-service	0.6303	0.9532	0.8182	0.0659	0.0050	0.0277	0.0128	0.2957
Farming-fishing	0.6953	0.9166	0.6087	0.0607	0.0463	0.1080	0.0892	0.6168
Sales	0.7688	0.8638	0.4218	0.1664	0.0752	0.1069	0.2069	0.4006
Exec-managerial	0.7960	0.8469	0.2416	0.0934	0.1651	0.2146	0.4654	0.6901
Craft-repair	0.6848	0.8391	0.5578	0.1796	0.0985	0.1429	0.1746	0.4673
Prof-specialty	0.8007	0.8689	0.2641	0.0862	0.1398	0.1801	0.3960	0.6522
Machine-op-inspct	0.6293	0.9150	0.7500	0.0876	0.0388	0.0824	0.0702	0.4807
Tech-support	0.7973	0.8863	0.3697	0.1301	0.0691	0.0982	0.2386	0.4626
Adm-clerical	0.7528	0.8839	0.4840	0.1567	0.0473	0.0686	0.1097	0.2777
Missing	0.7236	0.8602	0.5942	0.2465	0.0263	0.0449	0.0629	0.1726
Handlers-cleaners	0.7215	0.9560	0.6389	0.0461	0.0126	0.0418	0.0370	0.5077
Transport-moving	0.6963	0.6956	0.5556	0.5479	0.0812	0.0919	0.1493	0.1775
Protective-serv	0.7113	0.8563	0.4478	0.0652	0.1420	0.2222	0.2620	0.7731
Priv-house-serv	0.7400	0.9832	0.6667	0.0043	0.0000	0.0204	0.0192	0.9020
Armed-Forces	1.0000	0.5790	0	0.0464	0.0000	0.0000	0.0000	0.9487

Relationship

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Own-child	0.7707	0.9735	0.6250	0.0669	0.0000	0.0037	0.0047	0.1367
Husband	0.7263	0.8162	0.3502	0.0878	0.2001	0.2904	0.4038	0.7438
Not-in-family	0.7354	0.8765	0.6290	0.2741	0.0095	0.0236	0.0451	0.1746
Unmarried	0.7087	0.9412	0.7059	0.1356	0.0024	0.0114	0.0214	0.1769
Wife	0.7121	0.7686	0.3282	0.2086	0.2484	0.2547	0.4397	0.5359
Other-relative	0.6685	0.9525	0.7857	0.0726	0.0000	0.0237	0.0085	0.4611

Race

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
White	0.7836	0.8870	0.3850	0.1136	0.0768	0.1123	0.2139	0.5020
Black	0.7899	0.9374	0.4792	0.0897	0.0211	0.0393	0.0818	0.3546
Amer-Indian-Eskimo	0.7143	0.8882	0.5455	0.1628	0.0435	0.0609	0.0794	0.2722
Asian-Pac-Islander	0.7697	0.8739	0.4167	0.1345	0.0769	0.1172	0.2087	0.5027
Other	0.6720	0.9489	0.7273	0.0210	0.0130	0.0519	0.0455	0.8499

Sex

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Female	0.7863	0.9372	0.4724	0.0944	0.0241	0.0369	0.0804	0.3576
Male	0.7729	0.8714	0.3779	0.1123	0.0985	0.1451	0.2571	0.5596

Native-country

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
United-States	0.7847	0.8898	0.3882	0.1134	0.0727	0.1069	0.2037	0.4891
Ireland	0.8125	0.8717	0.4000	0.0741	0.0000	0.1429	0.2500	0.7647
Cuba	0.8137	0.8879	0.3750	0.1333	0.0455	0.0909	0.2000	0.4054
Missing	0.8390	0.8140	0.3158	0.2836	0.0461	0.0987	0.2201	0.2877
Puerto-Rico	0.9231	0.9496	0.2500	0.0545	0.0000	0.0444	0.0612	0.5400
Philippines	0.7349	0.8830	0.5000	0.1111	0.0702	0.1228	0.1644	0.5250
China	0.7818	0.8348	0.3000	0.1333	0.1364	0.1818	0.3125	0.4595
Germany	0.8393	0.8623	0.3571	0.1750	0.0250	0.1000	0.1852	0.4625
Mexico	0.6026	0.9397	0.8333	0.1277	0.0098	0.0146	0.0142	0.1746
Peru	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000	0.0909	0.7872
India	0.7333	0.8250	0.4706	0.1667	0.0526	0.1579	0.2778	0.6000
Guatemala	1.0000	1.0000	0	0.0000	0.0000	0.0000	0.0000	0.3824
Taiwan	0.8661	0.9003	0.1429	0.0303	0.1250	0.1250	0.4667	0.8784
Vietnam	0.6167	0.8752	0.0000	0.0000	0.1818	0.1364	0.2174	0.3448
Jamaica	0.4468	0.9626	1.0000	0.0571	0.0000	0.0000	0.0000	0.5893

Hungary	1.0000	0.9158	0.0000	0.0500	0.0000	0.0000	0.2500	0.8261
Portugal	0.4118	0.8960	1.0000	0.0612	0.0000	0.0000	0.0000	0.8214
Greece	0.6410	0.7879	0.5000	0.0909	0.1667	0.3333	0.3571	0.7857
Haiti	0.8039	0.9333	0.3333	0.0625	0.0588	0.0588	0.1500	0.6327
Nicaragua	1.0000	0.9544	0	0.0690	0.0000	0.0000	0.0000	0.5870
South	0.8462	0.9637	0.2857	0.0200	0.0526	0.0526	0.2308	0.7246
France	0.4167	0.6042	0.3333	0.0667	0.7500	0.7500	0.7143	0.8947
Canada	0.6485	0.9165	0.6471	0.1034	0.0645	0.0645	0.1667	0.4667
El-Salvador	0.9206	0.9344	0.2500	0.1818	0.0000	0.0000	0.0857	0.2143
England	0.7039	0.8390	0.5385	0.2400	0.0800	0.0800	0.2105	0.4200
Poland	0.5567	1.0000	0.8333	0.0000	0.0000	0.0000	0.0556	0.8095
Outlying-US (Guam-USV I-etc)	1.0000	1.0000	0	0.0000	0.0000	0.0000	0.0000	0.7353
Honduras	0.4706	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000	0.7037
Dominican- Republic	0.8857	0.9365	0.0000	0.0714	0.0556	0.0556	0.1500	0.4375
Italy	0.7375	0.8853	0.5000	0.1111	0.0556	0.1111	0.2143	0.5778
Trinidad&T obago	1.0000	0.9114	0.0000	0.0256	0.0000	0.1111	0.1000	0.8851
Iran	0.5238	0.7826	0.6667	0.0882	0.2727	0.3636	0.2857	0.7778
Columbia	0.7206	0.9685	0.0000	0.0000	0.1111	0.0556	0.1579	0.4688
Hong	0.4167	0.9394	1.0000	0.0000	0.1667	0.1667	0.1429	0.7727
Yugoslavia	0.4167	0.4545	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
Ecuador	1.0000	1.0000	0	0.0000	0.0000	0.0000	0.0000	0.4167
Cambodia	0.7083	0.8125	0.5000	0.2222	0.0000	0.0000	0.2857	0.5833
Japan	0.8873	0.9282	0.2857	0.0690	0.0000	0.0625	0.2174	0.6222
Scotland	1.0000	0.9396	0.0000	0.0192	0.0000	0.0000	0.2000	0.9107
Laos	0.7778	0.9372	0.5000	0.0286	0.0000	0.0000	0.1667	0.8718
Thailand	1.0000	0.9625	0	0.0000	0.0000	0.1250	0.0000	0.8971

Stage 6: Post-Mitigation ML Model (AIF360 Reweighting)

Detailed Group Metrics: Age

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Early-Middle	4582	0.8117	0.7700	0.3112	0.2634	0.3794	0.1020
Senior	890	0.8528	0.7821	0.2438	0.1865	0.4194	0.0594
Young	3990	0.9529	0.7294	0.0596	0.0316	0.6303	0.0101
Late-Middle	2749	0.7774	0.7576	0.3787	0.3358	0.3506	0.1446

Detailed Group Metrics: Education

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Bachelors	2071	0.8025	0.7953	0.4153	0.3974	0.2593	0.1536
9th	187	0.9679	0.4918	0.0321	0.0000	1.0000	0.0000
HS-grad	4011	0.8661	0.6997	0.1576	0.0980	0.6139	0.0441
Assoc-voc	497	0.8028	0.7261	0.2656	0.2052	0.4848	0.0932
Prof-school	224	0.8616	0.8187	0.7009	0.7857	0.0382	0.3731
10th	355	0.9408	0.7006	0.0789	0.0254	0.7143	0.0031
Some-college	2683	0.8550	0.7324	0.1826	0.1405	0.5122	0.0629
7th-8th	249	0.9197	0.5622	0.0884	0.0080	0.9091	0.0000
Masters	637	0.7849	0.7838	0.5589	0.5133	0.2331	0.1922
11th	442	0.9570	0.6824	0.0475	0.0226	0.7143	0.0095
5th-6th	99	0.9091	0.4762	0.0909	0.0000	1.0000	0.0000
Assoc-acdm	379	0.8179	0.7395	0.2427	0.2084	0.4457	0.0976
Preschool	22	1.0000	1.0000	0.0455	0.0455	0.0000	0.0000
12th	154	0.9481	0.7640	0.0844	0.0325	0.6154	0.0000
Doctorate	155	0.7419	0.6869	0.6581	0.7613	0.1176	0.5283
1st-4th	46	0.9783	0.4945	0.0217	0.0000	1.0000	0.0000

Detailed Group Metrics: Marital-status

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Never-married	4065	0.9697	0.7529	0.0440	0.0192	0.6257	0.0028

Married-civ-s pouse	5567	0.7315	0.7253	0.4489	0.4009	0.3525	0.2001
Divorced	1682	0.9298	0.7332	0.0957	0.0458	0.6273	0.0112
Separated	373	0.9491	0.7753	0.0777	0.0429	0.5517	0.0087
Married-spou se-absent	152	0.9145	0.7358	0.1184	0.0592	0.6111	0.0149
Widowed	359	0.9387	0.7088	0.0864	0.0251	0.7097	0.0000
Married-AF-s pouse	13	0.6923	0.5667	0.3846	0.0769	0.8000	0.0000

Detailed Group Metrics: Occupation

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Other-service	1248	0.9599	0.6326	0.0441	0.0120	0.8182	0.0042
Farming-fishi ng	370	0.8784	0.6947	0.1243	0.1000	0.5870	0.0556
Sales	1368	0.8282	0.7637	0.2617	0.2156	0.4162	0.0851
Exec-manage rial	1472	0.7955	0.7954	0.5061	0.4701	0.2376	0.1706
Craft-repair	1564	0.8050	0.6898	0.2212	0.1694	0.5578	0.0920
Prof-specialty	1568	0.8099	0.8041	0.4298	0.3967	0.2596	0.1376
Machine-op-i nspct	727	0.8556	0.6209	0.1486	0.0646	0.7685	0.0355
Tech-support	394	0.8173	0.7663	0.3020	0.2310	0.4202	0.0800
Adm-clerical	1413	0.8946	0.7540	0.1331	0.1111	0.4787	0.0482
Missing	715	0.9231	0.7314	0.0965	0.0587	0.5942	0.0217
Handlers-clea ners	514	0.9533	0.7569	0.0700	0.0311	0.6111	0.0042
Transport-mo ving	576	0.8194	0.6750	0.1875	0.1458	0.5926	0.0855
Protective-ser v	229	0.7817	0.7235	0.2926	0.2489	0.4478	0.1235
Priv-house-se rv	52	0.9808	0.8949	0.0577	0.0385	0.3333	0.0000
Armed-Force s	1	1.0000	1.0000	0.0000	0.0000	0	0.0000

Detailed Group Metrics: Relationship

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Own-child	1904	0.9916	0.7626	0.0126	0.0053	0.6250	0.0005
Husband	4916	0.7299	0.7241	0.4530	0.4022	0.3543	0.2004
Not-in-family	3148	0.9314	0.7467	0.0985	0.0476	0.6065	0.0099
Unmarried	1311	0.9512	0.7065	0.0648	0.0221	0.7059	0.0033
Wife	580	0.7259	0.7224	0.4517	0.4362	0.3206	0.2358
Other-relative	352	0.9688	0.6685	0.0398	0.0085	0.7857	0.0000

Detailed Group Metrics: Race

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
White	10442	0.8448	0.7836	0.2547	0.2137	0.3853	0.0766
Black	1186	0.9216	0.7836	0.1214	0.0801	0.4931	0.0211
Amer-Indian-Eskimo	126	0.9206	0.7510	0.0873	0.0873	0.4545	0.0435
Asian-Pac-Islander	369	0.8347	0.7733	0.2602	0.2195	0.3958	0.0842
Other	88	0.8977	0.6720	0.1250	0.0455	0.7273	0.0130

Detailed Group Metrics: Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Female	4057	0.9275	0.7915	0.1117	0.0806	0.4636	0.0233
Male	8154	0.8160	0.7718	0.3028	0.2569	0.3795	0.0990

Detailed Group Metrics: Native-country

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
United-States	10993	0.8507	0.7849	0.2430	0.2038	0.3879	0.0727
Ireland	12	0.8333	0.8125	0.4167	0.2500	0.4000	0.0000
Cuba	30	0.8667	0.8137	0.2667	0.2000	0.3750	0.0455
Missing	209	0.8660	0.8231	0.2727	0.2344	0.3158	0.0658
Puerto-Rico	49	0.9796	0.9231	0.0816	0.0612	0.2500	0.0000
Philippines	73	0.8493	0.7633	0.2192	0.1781	0.4375	0.0702

China	32	0.8750	0.8454	0.3125	0.2500	0.3000	0.0455
Germany	54	0.8519	0.7857	0.2593	0.1852	0.4286	0.0500
Mexico	211	0.9668	0.6026	0.0284	0.0142	0.8333	0.0098
Peru	11	0.9091	0.8070	0.0909	0.1818	0.0000	0.1000
India	36	0.6944	0.6824	0.4722	0.3333	0.4706	0.1579
Guatemala	21	1.0000	1.0000	0.0000	0.0000	0	0.0000
Taiwan	15	0.8667	0.8661	0.4667	0.4667	0.1429	0.1250
Vietnam	23	0.9130	0.7262	0.0435	0.1304	0.0000	0.0909
Jamaica	26	0.8077	0.4468	0.1923	0.0000	1.0000	0.0000
Hungary	4	0.7500	0.4286	0.2500	0.0000	1.0000	0.0000
Portugal	10	0.7000	0.4118	0.3000	0.0000	1.0000	0.0000
Greece	14	0.6429	0.6410	0.5714	0.3571	0.5000	0.1667
Haiti	20	0.9500	0.8857	0.1500	0.1000	0.3333	0.0000
Nicaragua	17	1.0000	1.0000	0.0000	0.0000	0	0.0000
South	26	0.8846	0.8462	0.2692	0.2308	0.2857	0.0526
France	7	0.4286	0.4167	0.4286	0.7143	0.3333	0.7500
Canada	48	0.7708	0.7025	0.3542	0.1667	0.5882	0.0323
El-Salvador	35	0.9714	0.9206	0.1143	0.0857	0.2500	0.0000
England	38	0.7632	0.7039	0.3421	0.2105	0.5385	0.0800
Poland	18	0.7222	0.5567	0.3333	0.0556	0.8333	0.0000
Outlying-US(Guam-USVI-e tc)	9	1.0000	1.0000	0.0000	0.0000	0	0.0000
Honduras	9	0.8889	0.4706	0.1111	0.0000	1.0000	0.0000
Dominican-R epublic	20	0.9000	0.7222	0.1000	0.1000	0.5000	0.0556
Italy	28	0.7857	0.7375	0.3571	0.2143	0.5000	0.0556
Trinidad&To bago	10	1.0000	1.0000	0.1000	0.1000	0.0000	0.0000
Iran	14	0.6429	0.5238	0.2143	0.2857	0.6667	0.2727
Columbia	19	0.9474	0.8190	0.0526	0.1053	0.0000	0.0556
Hong	7	0.7143	0.4167	0.1429	0.1429	1.0000	0.1667
Yugoslavia	7	0.7143	0.4167	0.2857	0.0000	1.0000	0.0000
Ecuador	7	1.0000	1.0000	0.0000	0.0000	0	0.0000

Cambodia	7	0.7143	0.7083	0.5714	0.2857	0.5000	0.0000
Japan	23	0.9130	0.8873	0.3043	0.2174	0.2857	0.0000
Scotland	5	1.0000	1.0000	0.2000	0.2000	0.0000	0.0000
Laos	6	0.8333	0.7778	0.3333	0.1667	0.5000	0.0000
Thailand	8	1.0000	1.0000	0.0000	0.0000	0	0.0000

Detailed Group Metrics: Race + Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
White + Female	3261	0.9233	0.7953	0.1202	0.0889	0.4490	0.0258
Black + Male	600	0.8883	0.7780	0.1783	0.1167	0.4860	0.0304
White + Male	7181	0.8091	0.7696	0.3158	0.2703	0.3743	0.1062
Black + Female	586	0.9556	0.7786	0.0631	0.0427	0.5135	0.0128
Amer-Indian-Eskimo + Male	73	0.9315	0.7889	0.0685	0.1096	0.2000	0.0588
Asian-Pac-Islander + Female	131	0.9160	0.7568	0.1221	0.0687	0.5625	0.0174
Amer-Indian-Eskimo + Female	53	0.9057	0.6964	0.1132	0.0566	0.6667	0.0213
Asian-Pac-Islander + Male	238	0.7899	0.7584	0.3361	0.3025	0.3625	0.1329
Other + Female	26	0.9231	0.4800	0.0769	0.0000	1.0000	0.0000
Other + Male	62	0.8871	0.6992	0.1452	0.0645	0.6667	0.0189

Detailed Group Metrics: Age + Sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Early-Middle + Female	1396	0.8947	0.7991	0.1726	0.1375	0.4066	0.0424
Early-Middle + Male	3186	0.7753	0.7515	0.3719	0.3186	0.3738	0.1364
Senior + Male	609	0.8079	0.7634	0.3169	0.2496	0.4093	0.0913
Young + Male	2376	0.9419	0.7328	0.0753	0.0400	0.6201	0.0123

Late-Middle + Female	766	0.8916	0.7789	0.1684	0.1175	0.4729	0.0345
Young + Female	1614	0.9690	0.7143	0.0366	0.0192	0.6610	0.0071
Senior + Female	281	0.9502	0.8024	0.0854	0.0498	0.5000	0.0078
Late-Middle + Male	1983	0.7332	0.7293	0.4599	0.4201	0.3333	0.2101

Comparative Group Metrics: Before vs After (AIF360 Reweighing)

Age

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Early-Middle	0.7714	0.7700	0.3745	0.3794	0.1033	0.1020	0.2658	0.2634
Senior	0.7704	0.7821	0.4424	0.4194	0.0609	0.0594	0.1820	0.1865
Young	0.7329	0.7294	0.6176	0.6303	0.0112	0.0101	0.0333	0.0316
Late-Middle	0.7584	0.7576	0.3554	0.3506	0.1393	0.1446	0.3307	0.3358

Education

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Bachelors	0.7935	0.7953	0.2663	0.2593	0.1511	0.1536	0.3930	0.3974
9th	0.4918	0.4918	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
HS-grad	0.6985	0.6997	0.6108	0.6139	0.0468	0.0441	0.1007	0.0980
Assoc-voc	0.7353	0.7261	0.4697	0.4848	0.0904	0.0932	0.2072	0.2052
Prof-school	0.8341	0.8187	0.0446	0.0382	0.3284	0.3731	0.7679	0.7857
10th	0.6942	0.7006	0.7143	0.7143	0.0061	0.0031	0.0282	0.0254
Some-college	0.7370	0.7324	0.5082	0.5122	0.0602	0.0629	0.1390	0.1405
7th-8th	0.5622	0.5622	0.9091	0.9091	0.0000	0.0000	0.0080	0.0080
Masters	0.7914	0.7838	0.2191	0.2331	0.1922	0.1922	0.5212	0.5133
11th	0.6895	0.6824	0.7143	0.7143	0.0071	0.0095	0.0204	0.0226
5th-6th	0.4762	0.4762	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
Assoc-acdm	0.7194	0.7395	0.4783	0.4457	0.1045	0.0976	0.2058	0.2084
Preschool	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000	0.0455	0.0455

12th	0.7640	0.7640	0.6154	0.6154	0.0000	0.0000	0.0325	0.0325
Doctorate	0.6611	0.6869	0.1275	0.1176	0.5660	0.5283	0.7677	0.7613
1st-4th	0.4945	0.4945	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000

Marital-status

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Never-married	0.7459	0.7529	0.6369	0.6257	0.0031	0.0028	0.0189	0.0192
Married-civil-spouse	0.7259	0.7253	0.3501	0.3525	0.2011	0.2001	0.4026	0.4009
Divorced	0.7200	0.7332	0.6584	0.6273	0.0092	0.0112	0.0410	0.0458
Separated	0.7585	0.7753	0.5862	0.5517	0.0087	0.0087	0.0402	0.0429
Married-spouse-absent	0.7764	0.7358	0.5556	0.6111	0.0075	0.0149	0.0592	0.0592
Widowed	0.7088	0.7088	0.7097	0.7097	0.0000	0.0000	0.0251	0.0251
Married-AF-spouse	0.7068	0.5667	0.6000	0.8000	0.0000	0.0000	0.1538	0.0769

Occupation

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Other-service	0.6303	0.6326	0.8182	0.8182	0.0050	0.0042	0.0128	0.0120
Farming-fishing	0.6953	0.6947	0.6087	0.5870	0.0463	0.0556	0.0892	0.1000
Sales	0.7688	0.7637	0.4218	0.4162	0.0752	0.0851	0.2069	0.2156
Exec-managerial	0.7960	0.7954	0.2416	0.2376	0.1651	0.1706	0.4654	0.4701
Craft-repair	0.6848	0.6898	0.5578	0.5578	0.0985	0.0920	0.1746	0.1694
Prof-specialty	0.8007	0.8041	0.2641	0.2596	0.1398	0.1376	0.3960	0.3967
Machine-operator-inspct	0.6293	0.6209	0.7500	0.7685	0.0388	0.0355	0.0702	0.0646
Tech-support	0.7973	0.7663	0.3697	0.4202	0.0691	0.0800	0.2386	0.2310
Adm-clerical	0.7528	0.7540	0.4840	0.4787	0.0473	0.0482	0.1097	0.1111
Missing	0.7236	0.7314	0.5942	0.5942	0.0263	0.0217	0.0629	0.0587

Handlers-cleaners	0.7215	0.7569	0.6389	0.6111	0.0126	0.0042	0.0370	0.0311
Transport-moving	0.6963	0.6750	0.5556	0.5926	0.0812	0.0855	0.1493	0.1458
Protective-serv	0.7113	0.7235	0.4478	0.4478	0.1420	0.1235	0.2620	0.2489
Priv-house-serv	0.7400	0.8949	0.6667	0.3333	0.0000	0.0000	0.0192	0.0385
Armed-Forces	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000

Relationship

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Own-child	0.7707	0.7626	0.6250	0.6250	0.0000	0.0005	0.0047	0.0053
Husband	0.7263	0.7241	0.3502	0.3543	0.2001	0.2004	0.4038	0.4022
Not-in-family	0.7354	0.7467	0.6290	0.6065	0.0095	0.0099	0.0451	0.0476
Unmarried	0.7087	0.7065	0.7059	0.7059	0.0024	0.0033	0.0214	0.0221
Wife	0.7121	0.7224	0.3282	0.3206	0.2484	0.2358	0.4397	0.4362
Other-relative	0.6685	0.6685	0.7857	0.7857	0.0000	0.0000	0.0085	0.0085

Race

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
White	0.7836	0.7836	0.3850	0.3853	0.0768	0.0766	0.2139	0.2137
Black	0.7899	0.7836	0.4792	0.4931	0.0211	0.0211	0.0818	0.0801
Amer-Indian-Eskimo	0.7143	0.7510	0.5455	0.4545	0.0435	0.0435	0.0794	0.0873
Asian-Pac-Islander	0.7697	0.7733	0.4167	0.3958	0.0769	0.0842	0.2087	0.2195
Other	0.6720	0.6720	0.7273	0.7273	0.0130	0.0130	0.0455	0.0455

Sex

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Female	0.7863	0.7915	0.4724	0.4636	0.0241	0.0233	0.0804	0.0806
Male	0.7729	0.7718	0.3779	0.3795	0.0985	0.0990	0.2571	0.2569

Native-country

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
United-States	0.7847	0.7849	0.3882	0.3879	0.0727	0.0727	0.2037	0.2038
Ireland	0.8125	0.8125	0.4000	0.4000	0.0000	0.0000	0.2500	0.2500
Cuba	0.8137	0.8137	0.3750	0.3750	0.0455	0.0455	0.2000	0.2000
Missing	0.8390	0.8231	0.3158	0.3158	0.0461	0.0658	0.2201	0.2344
Puerto-Rico	0.9231	0.9231	0.2500	0.2500	0.0000	0.0000	0.0612	0.0612
Philippines	0.7349	0.7633	0.5000	0.4375	0.0702	0.0702	0.1644	0.1781
China	0.7818	0.8454	0.3000	0.3000	0.1364	0.0455	0.3125	0.2500
Germany	0.8393	0.7857	0.3571	0.4286	0.0250	0.0500	0.1852	0.1852
Mexico	0.6026	0.6026	0.8333	0.8333	0.0098	0.0098	0.0142	0.0142
Peru	1.0000	0.8070	0.0000	0.0000	0.0000	0.1000	0.0909	0.1818
India	0.7333	0.6824	0.4706	0.4706	0.0526	0.1579	0.2778	0.3333
Guatemala	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000
Taiwan	0.8661	0.8661	0.1429	0.1429	0.1250	0.1250	0.4667	0.4667
Vietnam	0.6167	0.7262	0.0000	0.0000	0.1818	0.0909	0.2174	0.1304
Jamaica	0.4468	0.4468	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
Hungary	1.0000	0.4286	0.0000	1.0000	0.0000	0.0000	0.2500	0.0000
Portugal	0.4118	0.4118	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
Greece	0.6410	0.6410	0.5000	0.5000	0.1667	0.1667	0.3571	0.3571
Haiti	0.8039	0.8857	0.3333	0.3333	0.0588	0.0000	0.1500	0.1000
Nicaragua	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000
South	0.8462	0.8462	0.2857	0.2857	0.0526	0.0526	0.2308	0.2308
France	0.4167	0.4167	0.3333	0.3333	0.7500	0.7500	0.7143	0.7143
Canada	0.6485	0.7025	0.6471	0.5882	0.0645	0.0323	0.1667	0.1667
El-Salvador	0.9206	0.9206	0.2500	0.2500	0.0000	0.0000	0.0857	0.0857
England	0.7039	0.7039	0.5385	0.5385	0.0800	0.0800	0.2105	0.2105
Poland	0.5567	0.5567	0.8333	0.8333	0.0000	0.0000	0.0556	0.0556
Outlying-US (Guam-USV l-etc)	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000
Honduras	0.4706	0.4706	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000

Dominican-Republic	0.8857	0.7222	0.0000	0.5000	0.0556	0.0556	0.1500	0.1000
Italy	0.7375	0.7375	0.5000	0.5000	0.0556	0.0556	0.2143	0.2143
Trinidad&Tobago	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000	0.1000	0.1000
Iran	0.5238	0.5238	0.6667	0.6667	0.2727	0.2727	0.2857	0.2857
Columbia	0.7206	0.8190	0.0000	0.0000	0.1111	0.0556	0.1579	0.1053
Hong	0.4167	0.4167	1.0000	1.0000	0.1667	0.1667	0.1429	0.1429
Yugoslavia	0.4167	0.4167	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000
Ecuador	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000
Cambodia	0.7083	0.7083	0.5000	0.5000	0.0000	0.0000	0.2857	0.2857
Japan	0.8873	0.8873	0.2857	0.2857	0.0000	0.0000	0.2174	0.2174
Scotland	1.0000	1.0000	0.0000	0.0000	0.0000	0.0000	0.2000	0.2000
Laos	0.7778	0.7778	0.5000	0.5000	0.0000	0.0000	0.1667	0.1667
Thailand	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000